

# Dominic Catania

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## OBJECTIVE

**Junior Computer Engineering student seeking an internship focused on data-driven systems and emerging technologies. Experienced in transforming system data into actionable insights.**

## EDUCATION

- Texas A&M University** College Station, TX  
*B.S. Computer Engineering; University Honors; GPA: 3.72* Aug 2024 – May 2028
  - Relevant Coursework: Data Structures & Algorithms; Discrete Structures for Computing; Statistics Principles

## FEATURED PROJECTS

- Depth Sensor and Turtle Tracker** May 2026 - Present  
*Tortugas Costa Rica - Sea Turtle Conservation Research*
  - Engineered a low-power marine telemetry system for sea turtle research using ESP32-based embedded hardware
  - Integrated high-pressure depth sensors, RTC synchronization, SD-card logging, and environmental sensing for long-duration underwater deployments with Iridium satellite transmission of oceanographic telemetry
- Red Light Green Light** October 2025  
*Texas A&M Datathon*
  - Designed automated data extraction pipelines using Python and Selenium to analyze dynamic system states
  - Applied rapid experimentation and iteration in a competitive, multidisciplinary environment
- Vigil** Jul 2024 - May 2025  
*FIRST Robotics Competition Team 4206*
  - Built a data-driven web platform analyzing robot performance and detecting anomalies with AI-assisted insights
  - Utilized Plotly and Tableau dashboards to support technical decision-making and performance optimization

## LEADERSHIP EXPERIENCE

- Triangle Fraternity** Oct 2024 - Present  
*Vice President of External Relations*
  - Built partnerships with campus organizations and coordinated inter-organizational events
  - Organized networking events with university leaders and engineering industry professionals
  - Coordinated Health & Safety, Public Relations, and Recruitment initiatives to support chapter operations
- FIRST Robotics Competition Team 4206** Aug 2021 - May 2024  
*Team Captain; Driver; Lead Programmer*
  - Led a 30+ member robotics team, overseeing project management, system integration, and collaboration
  - Placed 3rd in the World by optimizing performance through data logging and match strategy
  - Mentored 8 newer members in sensor integration, vision processing, and feedback systems enhancing robot precision
  - Incorporated distance interpolation, auto-aim, autonomous decision-making, full-field localization and pose estimation using inverse kinematics and AprilTag IDs, increasing robot performance by 80% over season

## ACHIEVEMENTS

- FIRST Robotics Championship - World Semifinalists**
- Reta & HJ Haynes '46 University Scholarship**
- Datathon Second Place**

## SKILLS

- Languages:** C++, Python, R, SQL, Java
- Technologies:** Tableau, Excel, Arduino, Power BI, Django